



**POLITECNICO**  
MILANO 1863

# Apache Spark

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# Exercise 1

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- Complete the bank account exercise
- Consider a new query Q4
  - Print all accounts in descending order of balance

# Exercise 2

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- Write a program that takes in input a csv file with the friendship relation in a social network
  - Two fields (user, friend)
- Write an iterative program that computes the transitive closure of the relation

# Exercise 3

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- Change the event enrichment program
  - Use the rate source to define the input stream
- Change the query to count the number of products of each class appearing in the input stream within a window of 30 seconds sliding every 5 seconds

# Eval lab 2022: assumptions

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## Three input datasets

### 1. citiesRegion

- Type: static, csv file
- Fields: city, region

### 2. citiesPopulation

- Type: static, csv file
- Fields: id (of the city), city, population

### 3. bookings

- Type: dynamic, stream
- Fields: timestamp, value
- Each entry with value x indicates that someone booked a hotel in the city with id x

# Eval lab 2022: requirements

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- For all queries: limit unnecessary recomputations as much as possible!
- Q1: compute the total population for each region
- Q2: compute the number of cities and the population of the most populated city for each region

# Eval lab 2022: requirements

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- Q3: Print the evolution of the population in Italy year by year until the total population in Italy overcomes 100M people
  - Assume that the population evolves as follows:
    - In cities with more than 1000 inhabitants, it increases by 1% every year
    - In cities with less than 1000 inhabitants, it decreased by 1% every year
  - The output on the terminal should be a sequence of lines
    - Year: 1, total population: xxx
    - Year: 2, total population: yyy
    - ...
  - You may round the population of each city to the nearest integer during your computation
- Q4: compute the total number of bookings for each region, in a window of 30 seconds, sliding every 5 seconds